MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

CAPACITOR, DIGITAL, KEY CODE FLE

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic capacitor module.

* 1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN. ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5963

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.5 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be FLE.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 75 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/127-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Preconditioning. Preconditioning is not required. However, the capacitors may be preconditioned by operating the module as shown on figure 2 for 24 hours at 80°C (+5°C, -0°C) Alternate preconditioning methods may be employed as specified in 4.4.1

3.10 Storage and handling. Modules shall not exhibit capacitance, dissipation, or rise times which exceed the 25°C ambient end-of-life requirements of table I for test conditions D, E, and F3-F18, respectively, due to nonoperating environments ≤85°C during storage and handling. The modules may be returned to within the 25°C initial limits of table I by preconditioning in the operating circuit shown on figure 2 for 24 hours at 80°C (+5°C, -0°C). Alternate preconditioning methods may be employed as specified in 4.4.1.

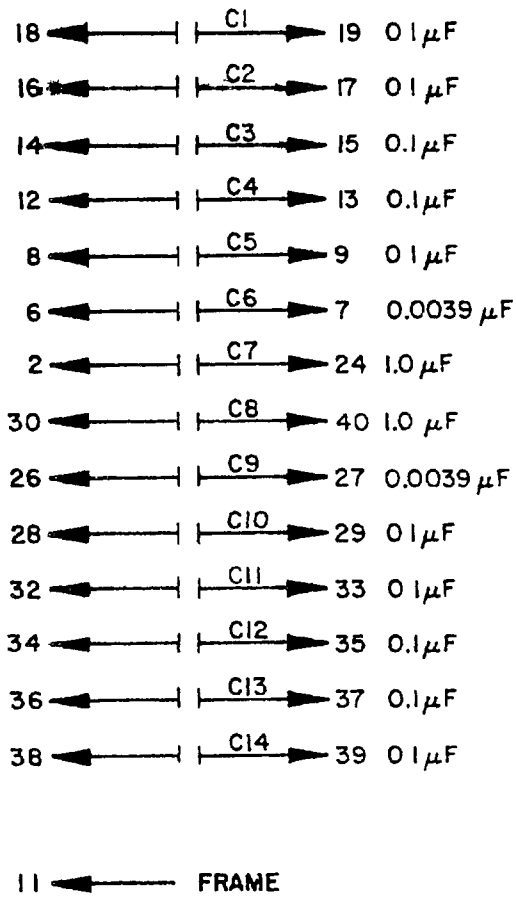
4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film unless otherwise specified.

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NOTES
1. Pins 1, 3, 4, 5, 10, 20, 21, 22, 23, 25, and 31 are not used
2. All capacitors are ±10%

FIGURE 1. FLE module function diagram, capacitor

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TABLE I Module electrical requirements

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A16	Isolation	II	X	10	∞	5	∞	10	∞	5	∞	MΩ
B1-B14	Insulation resistance	III	S	1000	∞	500	∞	500	∞	100	∞	MΩ
C	Continuity	IV	X	0 0	1 0	0.0	1 0	0 0	1 0	0 0	1 0	Ω
D1-D10	Capacitance (1 kHz)	V	X	0 09	0 115	0 08	0 125	0 08	0 125	0 07	0 135	µF
D11, D12	Capacitance (1 kHz)	V	X	0 0035	0 0045	0 0031	0 0049	0 0031	0.0049	0.0027	0.0053	µF
D13, D14	Capacitance (1 kHz)	V	X	0 9	1 15	0 8	1 25	0 8	1.25	0 7	1 35	µF
D15-D24	Capacitance (10 kHz)	V	S	0 085	0 115	0 075	0 125	0.075	0.125	0 065	0 135	µF
D25, D26	Capacitance (10 kHz)	V	S	0 0033	0 0045	0 0029	0 0049	0 0029	0 0049	0.0025	0 0053	µF
D27, D28	Capacitance (10 kHz)	V	S	0 85	1 15	0 75	1 25	0.75	1 25	0 65	1 35	µF
E1-E12	Dissipation factor (1 kHz)	VI	X	0 0	2 5	0 0	3 0	0 0	3 5	0 0	4 0	%
E13, E14	Dissipation factor (1 kHz)	VI	X	0 0	3 5	0 0	4 0	0 0	4 5	0 0	5 0	%
E15-E26	Dissipation factor (10 kHz)	VI	S	0 0	3 5	0 0	4 0	0 0	5 5	0 0	6.0	%
E27, E28	Dissipation factor (10 kHz)	VI	S	0 0	4 5	0 0	5 0	0 0	6 5	0 0	7 0	%
F1, F2	Rise time (parallel)	VII	A	260	300	---	---	---	---	---	---	µs
F3, F4	Rise time (parallel)	VII	X	260	339	245	356	248	353	230	370	µs
F5-F14	Rise time	VII	S	46	65	43	68	44	67	38	73	µs
F15, F16	Rise time	VII	S	17	25	15	27	16	26	14	29	µs
F17, F18	Rise time	VII	S	225	345	205	365	210	360	180	390	µs

1/ "X" parameters shall be 100 percent tested during module acceptance tests
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "A" parameter limits to be used in place of the limits for test conditions F3 and F4 at the module supplier level only (see 6.6)

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4.4 Test procedures. The module shall be tested as specified in table I

4 4.1 Alternate test methods Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787

4 4.2 Order of testing. Test conditions A through F shall be performed in any order with the following exceptions If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows A, B, C where applicable, followed by the remaining tests in the sequence

4 5 Qualification testing Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions

4.5 1 Nonoperating temperature test This test shall be performed prior to any other qualification testing The modules shall be subjected to a nonoperating temperature of 85°C (+0°C, -5°C) for a period of 1 hour. The temperature gradient shall not exceed 1°C per minute. At the end of the 85°C temperature stabilization period, the modules shall be returned to 25°C ambient and allowed to stabilize for a maximum of 24 hours The modules shall meet the 25°C end-of-life requirements in table I for test conditions D, E, and F3-F18 Following the electrical tests, the capacitors shall be preconditioned by operating the modules as shown on figure 2 for 24 hours at 80°C (+5°C, -0°C) Alternate preconditioning methods may be employed as specified in 4 4.1.

4 5 2 Operating temperature tests The operating temperature tests shall be conducted in accordance with MIL-M-28787 Before starting these tests, test conditions A through E and F3-F18 in table I shall be performed in the order specified in 4 4 2 using the initial limits at 25° ±5°C ambient When electrical testing is called for at the end of each temperature stabilization period, test conditions C through E and F3-F18 shall be performed in the order specified in 4 4 2 using the initial limits for applicable operating temperature At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2 The module shall also be operated as shown in the test circuit on figure 2 during the temperature transitions from 25°C initial to 0°C and from 80°C transient thermal to 25°C.

4 5.3 Thermal shock The thermal shock test shall be conducted in accordance with MIL-M-28787 with the exception that the module shall be cycled between the temperatures of 0°C (+0°C, -5°C) and 60°C (+5°C, -0°C). Following thermal shock the capacitors may be preconditioned by operating the module as shown on figure 2 for 24 hours at 80°C (+5°C, -0°C) Alternate preconditioning methods may be employed as specified in 4 4.1

4 5 4 Vibration tests The vibration tests shall be conducted in accordance with MIL-M-28787 During the vibration tests, the module shall not be operated Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient

4 5.5 Life test The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2 Following the life test and while still at the maximum class temperature, electrical tests C through E and F3-F18 shall be performed in the order specified in 4 4 2 using the operating temperature end-of-life requirements of table I Following the return of the module to 25°C, electrical tests C through E and F3-F18 shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5 PACKAGING

5 1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6 NOTES

6 1 Definitions Definitions are as specified in MIL-STD-133.

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6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military Specification MIL-M-28787/127 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658716. Part number M28787/127-1 supersedes part numbers 2658716 and M28787/127.

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	1.25 W
Voltage rating (each capacitor)	50 V minimum
Capacitance	
C1-C5 and C10-C14	0.1 µF ±30 percent
C6 and C9	0.0039 µF ±30 percent
C7 and C8	1.0 µF ±30 percent

* 6.5 Module function. This module contains 14 capacitors. Each capacitor has both leads independently accessible at the module pins. All capacitors are nonpolar. The ceramic capacitors used in this module may exhibit nonpermanent changes ≤±20 percent of the nominal capacitance value when subjected to nonoperating thermal cycling in the range of 60° to 85°C. System users should exercise caution during storage and handling of this module prior to use in critical timing applications.

* 6.6 Test type A. Test type A (test conditions F1 and F2) is a module supplier test at 25° ±5°C only. Upon delivery to the module user, test conditions F3 and F4 are applicable in lieu of test conditions F1 and F2.

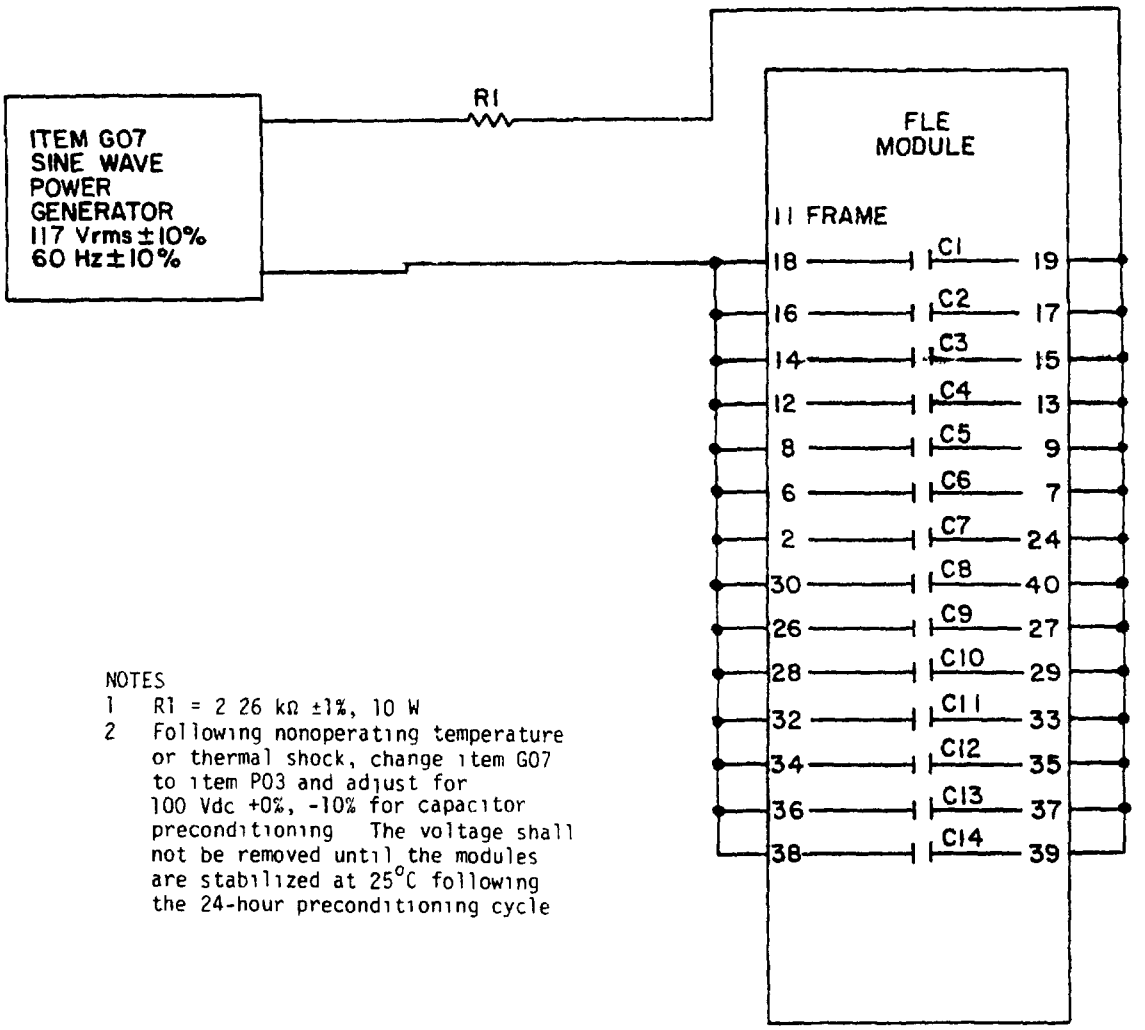
6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NOTES
1 R1 = 2 26 kΩ ±1%, 10 W
2 Following nonoperating temperature or thermal shock, change item G07 to item P03 and adjust for 100 Vdc +0%, -10% for capacitor preconditioning The voltage shall not be removed until the modules are stabilized at 25°C following the 24-hour preconditioning cycle

FIGURE 2 Operating temperature, life test, and preconditioning test circuit

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TABLE II Isolation test

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	1,3,4,5,10, 20,21,22,23, 25,31	2,6,7,8,9,11,12,13,14, 15,16,17,18,19,24,26, 27,28,29,30,32,33,34, 35,36,37,38,39,40	10	∞	MΩ	3
A2	C7	7,24	6,7,8,9,11,12,13,14, 15,16,17,18,19,26,27, 28,29,30,32,33,34,35, 36,37,38,39,40				
A3	C6	6,7	2,8,9,11,12,13,14,15, 16,17,18,19,24,26,27, 28,29,30,32,33,34,35, 36,37,38,39,40				
A4	C5	8,9	2,6,7,11,12,13,14,15, 16,17,18,19,24,26,27, 28,29,30,32,33,34,35, 36,37,38,39,40				
A5		11	2,6,7,8,9,12,13,14,15, 16,17,18,19,24,26,27, 28,29,30,32,33,34,35, 36,37,38,39,40				
A6	C4	12,13	2,6,7,8,9,11,14,15,16, 17,18,19,24,26,27,28, 29,30,32,33,34,35,36, 37,38,39,40				
A7	C3	14,15	2,6,7,8,9,11,12,13,16, 17,18,19,24,26,27,28, 29,30,32,33,34,35,36, 37,38,39,40				

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TABLE II Isolation test - Continued

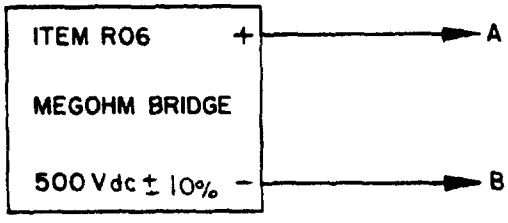
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A8	C2 Isolation	16,17	2,6,7,8,9,11,12,13,14, 15,18,19,24,26,27,28, 29,30,32,33,34,35,36, 37,38,39,40	10	∞	MΩ	3
A9	C1	18,19	2,6,7,8,9,11,12,13,14, 15,16,17,24,26,27,28, 29,30,32,33,34,35,36, 37,38,39,40				
A10	C9	26,27	2,6,7,8,9,11,12,13,14, 15,16,17,18,19,24,28, 29,30,32,33,34,35,36, 37,38,39,40				
A11	C10	28,29	2,6,7,8,9,11,12,13,14, 15,16,17,18,19,24,26, 27,30,32,33,34,35,36, 37,38,39,40				
A12	C8	30,40	2,6,7,8,9,11,12,13,14, 15,16,17,18,19,24,26, 27,28,29,32,33,34,35, 36,37,38,39				
A13	C11	32,33	2,6,7,8,9,11,12,13,14, 15,16,17,18,19,24,26, 27,28,29,30,34,35,36, 37,38,39,40				
A14	C12	34,35	2,6,7,8,9,11,12,13,14, 15,16,17,18,19,24,26, 27,28,29,30,32,33,36, 37,38,39,40				

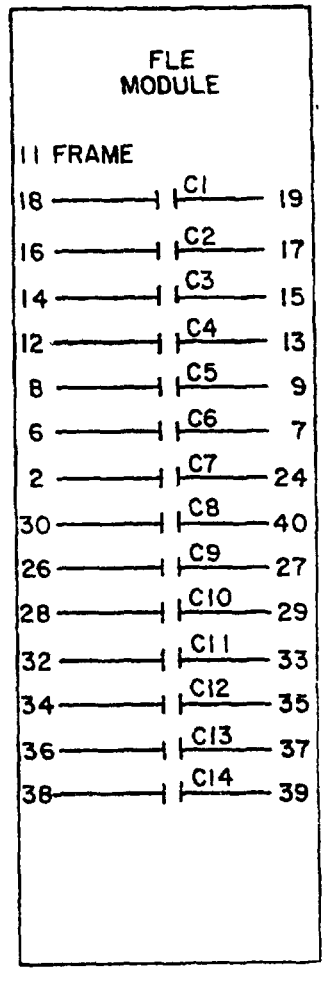
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TABLE II Isolation test - Continued

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A15	C13 Isolation	36,37	2,6,7,8,9,11,12,13,14, 15,16,17,18,19,24,26, 27,28,29,30,32,33,34, 35,38,39,40	10	∞	MΩ	3
A16	C14 Isolation	38,39	2,6,7,8,9,11,12,13,14, 15,16,17,18,19,24,26, 27,28,29,30,32,33,34, 35,36,37,40	10	∞	MΩ	3

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE Bridge voltage shall be applied for a minimum of 30 seconds prior to making measurement.

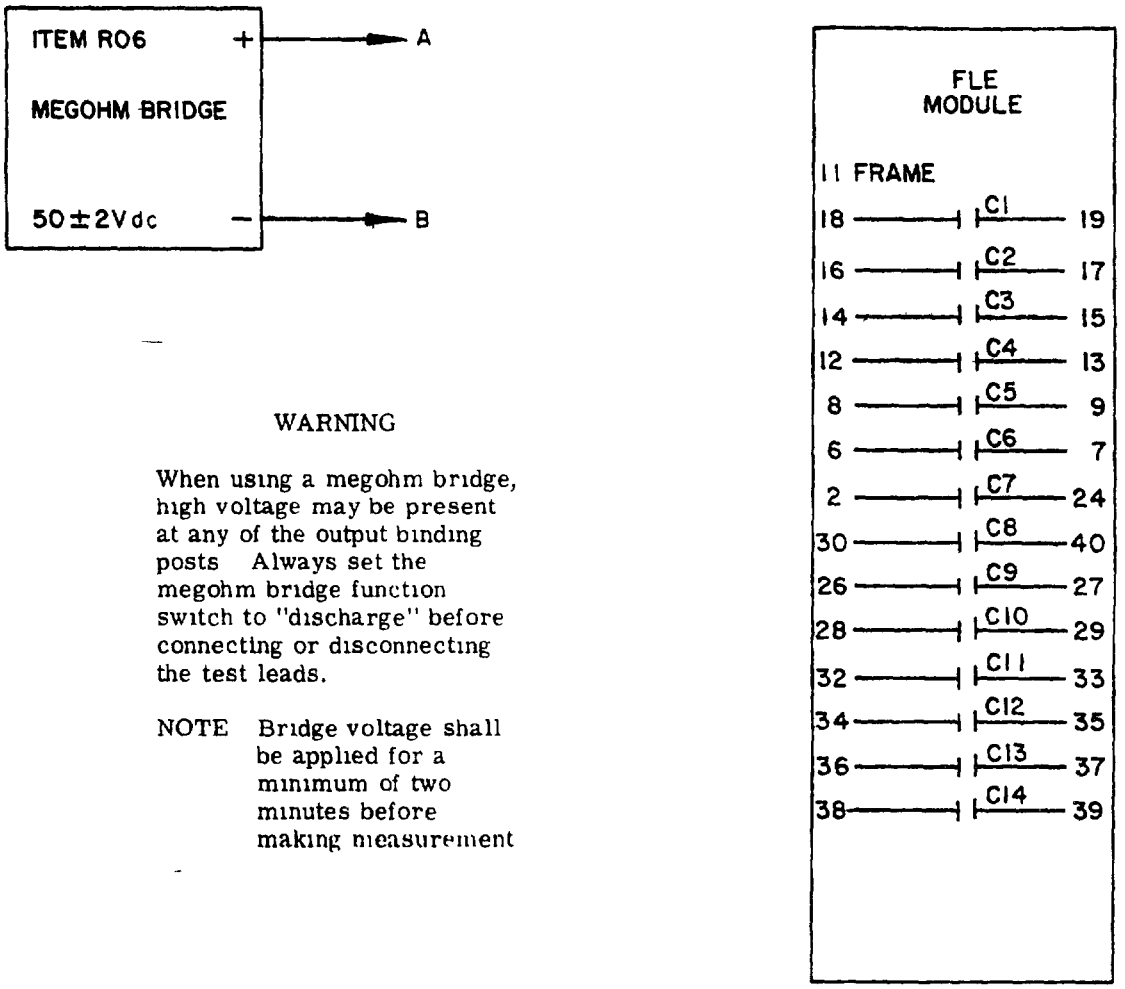
FIGURE 3 Isolation test circuit for table II.

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TABLE III Insulation resistance test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B1	C6	Insulation resistance	6	7	1000	∞	MΩ	4
B2	C9	↓	26	27	↓	↓	↓	↓
B3	C5		8	9				
B4	C4		12	13				
B5	C3		14	15				
B6	C2		16	17				
B7	C1		18	19				
B8	C10		28	29				
B9	C11		32	33				
B10	C12		34	35				
B11	C13		36	37				
B12	C14		38	39				
B13	C7		2	24				
B14	C8		30	40				

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE Bridge voltage shall be applied for a minimum of two minutes before making measurement

FIGURE 4 Insulation resistance test circuit for table III

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TABLE IV Continuity test

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C	Continuity	11	Frame	0 0	1 0	Ω	5

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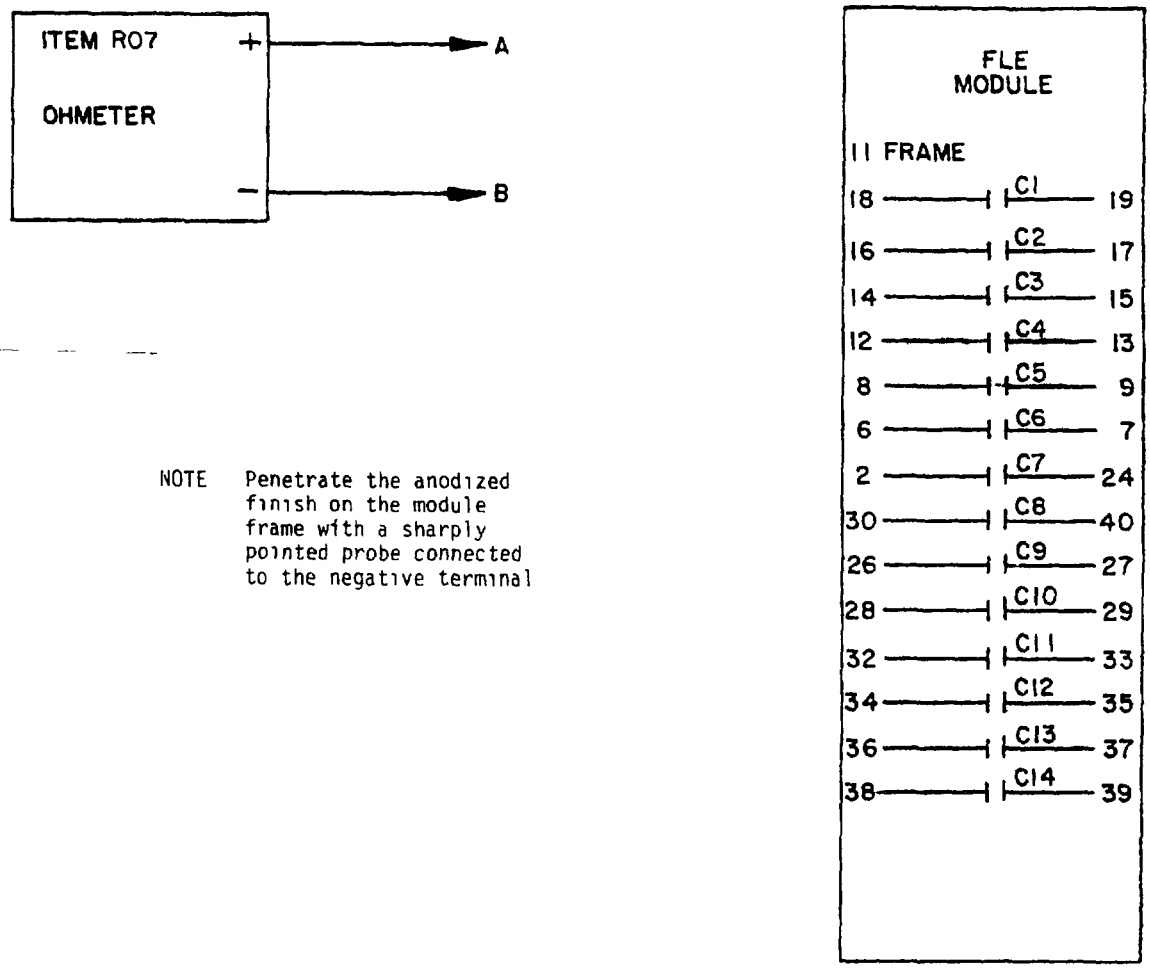


FIGURE 5. Continuity test circuit for table IV.

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TABLE V. Capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	CS Capacitance	8	9	0 09	0.115	µF	6
D2	C4	12	13				
D3	C3	14	15				
D4	C2	16	17				
D5	C1	18	19				
D6	C10	28	29				
D7	C11	32	33				
D8	C12	34	35				
D9	C13	36	37				
D10	C14	38	39				
D11	C6	6	7	0.0035	0.0045		
D12	C9	26	27	0.0035	0.0045		
D13	C7	2	24	0.9	1 15		
D14	C8	30	40	0.9	1.15		
D15	C5	8	9	0.085	0.115		
D16	C4	12	13				
D17	C3	14	15				
D18	C2	16	17				
D19	C1	18	19				

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TABLE V Capacitance test - Continued

Test condition	Parameter	Connect		Initial limits 25° +5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D20	C10 Capacitance	28	29	0 085	0 115	µF	6
D21	C11	32	33				
D22	C12	34	35				
D23	C13	36	37				
D24	C14	38	39				
D25	C6	6	7	0 0033	0 0045		
D26	C9	26	27	0 0033	0 0045		
D27	C7	2	24	0 85	1 15		
D28	C8	30	40	0 85	1 15		

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TABLE VI Dissipation factor test

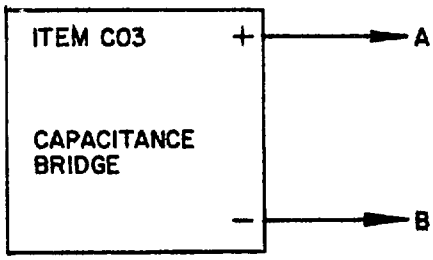
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	C5 Dissipation factor	8	9	0.0	2.5	%	6
E2	C4	12	13				
E3	C3	14	15				
E4	C2	16	17				
E5	C1	18	19				
E6	C10	28	29				
E7	C11	32	33				
E8	C12	34	35				
E9	C13	36	37				
E10	C14	38	39				
E11	C6	6	7				
E12	C9	26	27				
E13	C7	2	24		3.5		
E14	C8	30	40				
E15	C5	8	9				
E16	C4	12	13				
E17	C3	14	15				
E18	C2	16	17				
E19	C1	18	19				

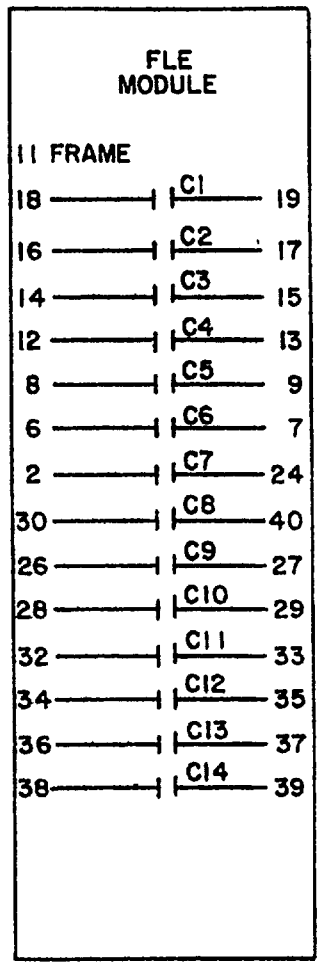
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TABLE VI Dissipation factor test - Continued

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E20	C10 Dissipation factor	28	29	0 0	3.5	%	6
E21	C11	32	33				
E22	C12	34	35				
E23	C13	36	37				
E24	C14	38	39				
E25	C6	6	7				
E26	C9	26	27				
E27	C7	2	24		4 5		
E28	C8	30	40		4 5		

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NOTES:

1. For test conditions D1-D14 and E1-E14, the C03 capacitance bridge output shall be set at:

   1.0 Vrms ±5%
   1.0 kHz ±10%

2. For test conditions D15-D28 and E15-E28, the C03 capacitance bridge output shall be set at:

   1.0 Vrms ±5%
   10.0 kHz ±5%

3. On the C03 capacitance bridge (GR1620 System) the following connections are necessary to obtain the proper test voltage:

   GR1311A audio oscillator output to GR1615A detector.

   GR1615A generator terminals to 1232A detector preamp input.

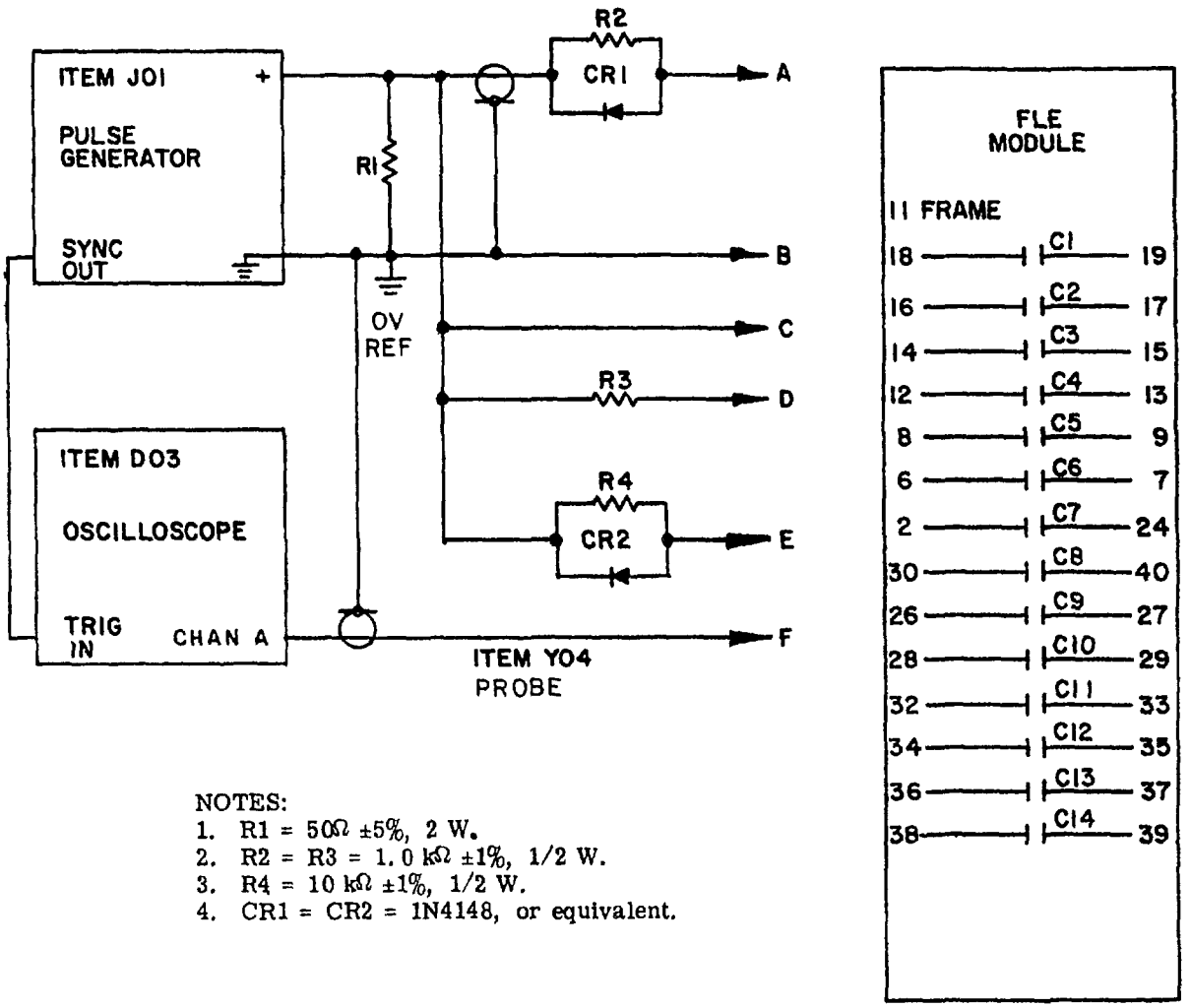
FIGURE 6 Capacitance and dissipation factor test circuit for tables V and VI.

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TABLE VII Rise time test

Test condition	Parameter	Connect						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
F1	C1-C6 Rise time (parallel)	7, 9, 13, 15, 17,19	6, 8, 12, 14, 16,18	Open	Open	Open	A	260	300	µs	7, 8
F2	C9-C14	27,29, 33,35, 37,39	26,28, 32,34, 36,38						300		
F3	C1-C6	7, 9, 13,15, 17,19	6, 8, 12,14, 16,18						339		
F4	C9-C14	27,29, 33,35, 37,39	26,28, 32,34, 36,38						339		
F5	C5 Rise time	9	8					46	65		
F6	C4	13	12								
F7	C3	15	14								
F8	C2	17	16								
F9	C1	19	18								
F10	C10	29	28								
F11	C11	33	32								
F12	C12	35	34								
F13	C13	37	36								
F14	C14	39	38								
F15	C6	Open	6			7	E	17	25		
F16	C9	Open	26			27	E	17	25		
F17	C7	D	2		24	Open	D	225	345		
F18	C8	D	30		40	Open	D	225	345		

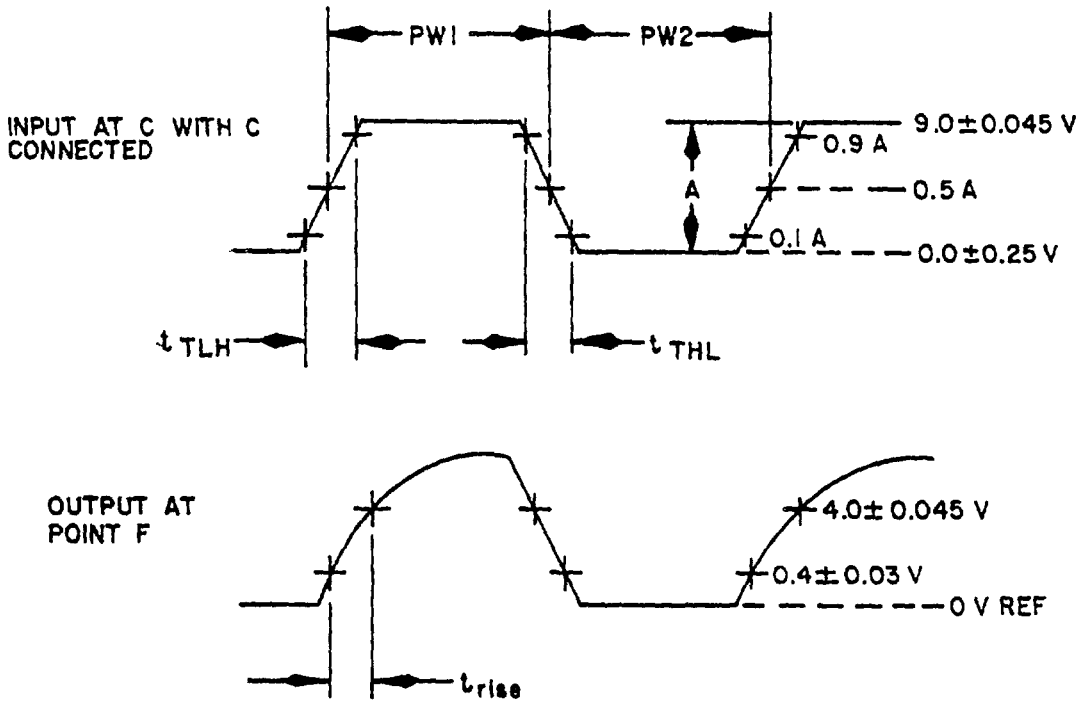
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NOTES:
1. R1 = 50Ω ±5%, 2 W.
2. R2 = R3 = 1.0 kΩ ±1%, 1/2 W.
3. R4 = 10 kΩ ±1%, 1/2 W.
4. CR1 = CR2 = 1N4148, or equivalent.

FIGURE 7. Rise time test circuit for table VII.

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NOTES
1 Input waveform parameters
  a. PW1 = PW2 = 3 0 ±0 15 ms
  b. $t_{TLH} = t_{THL} = 50 \pm 10$ ns
2 Rise time measurement must be made using the delay time multiplier decapot.

FIGURE 8 Rise time test circuit waveforms for table VII.

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